

Software Project Management (SPM) (CSC415)

Course Instructor
Er. ANUJ GHIMIRE

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Chapter 2: Project Analysis

Introduction

- Project analysis is the process of examining the aspects of a project in details.
- It helps to see that the project runs as expected and is also within the predefined budget.
- Its benefit includes:
 - *Determines Feasibility of a Project*
 - *Aids in Budgeting*
 - *Improves Project Planning and Scheduling*
 - *Detects and Mitigates Risks*
 - *Expedites the Monitoring and Evaluation of Projects*

Strategic Assessment (SA)

- In a strategic assessment, a successful assessment of a project is done so that there is a concrete plan that clearly defines an organizations objectives.
- It provides the basis for defining a project and it's goals.
- Strategic assessment is helpful in:
 - *Analyzing company performance*
 - *Identifying areas of strength and opportunity*
 - *Creating an workable environment.*
- Strategic assessment is beneficial to judge whether a project fits in the long-term goal of the organization.

Strategic Assessment (SA)

- It is usually approved out by senior management that needs strategic plan that clearly defines the objectives of the organization.
- Such assessment are done for:
 - *Programme Management*
 - *Portfolio Management*

SA_Programme Management

- Programme Management are suitable for projects developed for use in the organization.
- Individual projects are components of a programme management within the organization
- According to D.C. Ferns who is a journal of Project Management Institute *“Programme is “a group of projects that are managed in a coordinated way to gain benefits that would not be possible were the projects to be managed independently”.*

SA_Programme Management

- Issues of programme management are:
 - **Objectives:**
 - *How does the project contribute to the long-term goal of the organization?*
 - *Will the product increase the market share? And by how much?*
 - **IS (Integrated System) Plan**
 - *Does the product fit into the overall IS plan?*
 - *How does the product relate to other existing systems?*

SA_Programme Management

- **Organization structure**
 - *How does the product affect the existing organizational structure?*
 - *How does the product affect the existing the existing workflow?*
 - *What is the overall business model?*
- **MIS**
 - *What information does the product provide?*
 - *To whom is the information provided?*
 - *How does the product relate to other existing MIS's?*

SA_Programme Management

- **Personnel**

- *What are the staff implications?*
- *What are the impacts on the overall policy on staff development?*
 - *Staff implications includes skills and numbers*
 - *Staff development includes trainings, workshops, seminars, conferences, and magazine subscriptions etc.*

- **Good Will**

- *How does the product affect the good will of the organization?*

SA_Portfolio Management

- Suitable for product developed by a software company for an organization
- May need to assess the product for the client organization
- Need to carry out strategic assessment for the software providing company.
- Issues in Portfolio Management:
 - *Long-term goal of the software company*
 - *The effects of the project on the portfolio of the company (synergies and conflicts)*
 - *Any added-value to the overall portfolio of the company*

Technical Assessment (TA)

- Technical assessment of a project consists of evaluating the required functionality against the hardware and software available.
- It provides a fact-based understanding of the current level of product knowledge, technical maturity, program status and technical risk by comparing assessment results against defined criteria.
- Deals with:
 - *Functionality against hardware and software*
 - *The strategic IS plan of the organization*
 - *Any constraints imposed by the IS plan*

Technical Assessment (TA)

- Technical assessment is the crosscutting process used to help monitor technical progress of a project.
- To monitor the progress are done through periodic technical reviews and monitoring of technical indicators such as Key Performance Parameters (KPPs), and Technical Performance Measures (TPMs).

Economical Assessment (TA)

- Economic analysis in project analysis involves evaluating the project by analyzing financial feasibility of a project.
- This includes identifying and estimating costs of resources such as labor, materials, equipment, forecasting the project's revenue or savings, and assessing (i.e. judging) the potential risks and benefits of the of the project.
- The goal of economic analysis is to determine whether project is financially viable (i.e. feasible or workable) and to provide information that can be used to make decisions about the project's allocation of resources.

Economical Assessment (TA)

- There are several bases for comparing the worthiness of the projects.
- Some bases are:
 - *Present Worth Method*
 - *Future Worth Method*
 - *Annual Worth Method*
 - *Rate of return method*
 - *Internal Rate of Return (IRR) Analysis*
 - *Cost-Benefit Ratio (CBR) Analysis*

Present Worth Method

- It is also called Discounted Cash Flow (DCF) or Present value (PV) or Net Present Value (NPV).
- The NPV represents the difference between the present value of the cash inflows and the present value of cash outflows.
- In this method of comparison, the cash flows of each alternative will be reduced to time zero by assuming an interest rate i .
- If the NPV is positive, it indicates that the investment is expected to be a good investment and if NPV is negative, it means that the investment is not expected to be a good investment.

Present Worth Method

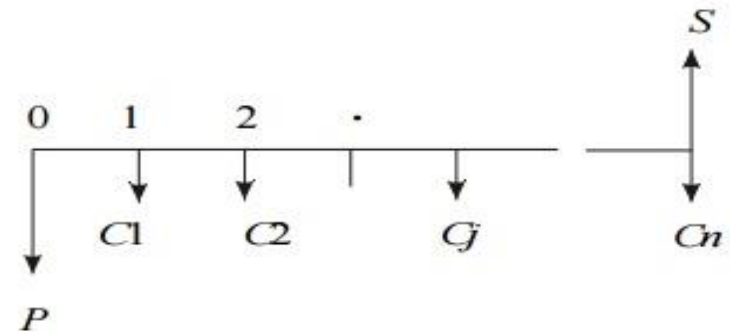
- Then, depending on the type of decision, the best alternative will be selected by comparing the present worth amounts of the alternatives.
- In a ***cost dominated cash flow diagram***, the costs (outflows) will be assigned with positive sign and the profit, revenue, salvage value (all inflows), etc. will be assigned with negative sign.
- In a ***revenue/profit-dominated cash flow diagram***, the profit, revenue, salvage value (all inflows to an organization) will be assigned with positive sign and the costs (outflows) will be assigned with negative sign.

Present Worth Method

- Here P represents an initial investment and R_j the net revenue at the end of the j^{th} year. The interest rate is i , compounded annually.
- S is the salvage value at the end of the w^{th} year.

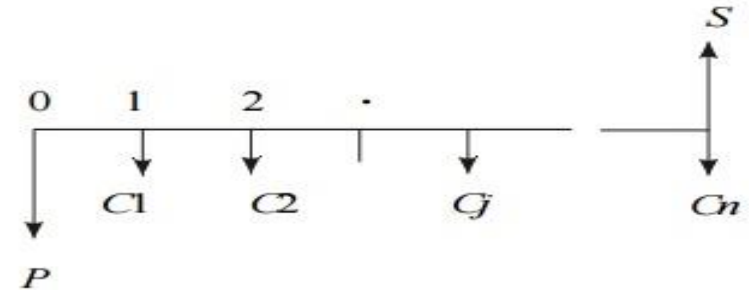
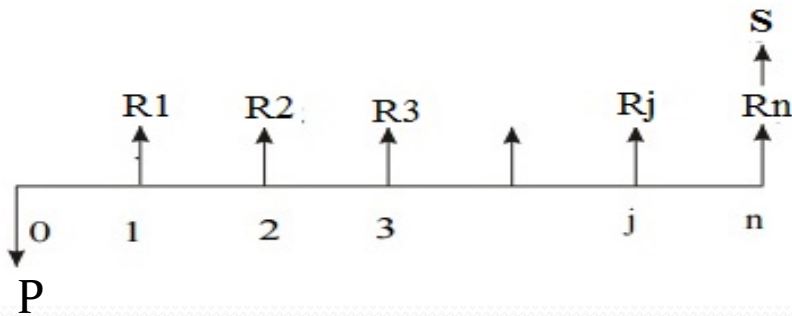


*Revenue/Profit-Dominated
Cash Flow Diagram*



*Cost-Dominated Cash
Flow Diagram*

Present Worth Method



- To find the present worth of the above cash flow diagram for a given interest rate, the formula is

RDCF $PW(i) = -P + R_1[1/(1+i)^1] + R_2[1/(1+i)^2] + \dots + R_j[1/(1+i)^j] + R_n[1/(1+i)] + S[1/(1+i)^n]$

CDCF $PW(i) = P + C_1[1/(1+i)^1] + C_2[1/(1+i)^2] + \dots + C_j[1/(1+i)^j] + C_n[1/(1+i)] - S[1/(1+i)^n]$

- In above formula for RDCF, expenditure is assigned a negative sign and revenues are assigned a positive sign but for CDCF, expenditure is assigned a positive sign and revenues are assigned a negative sign

Present Worth Method

- If we have some more alternatives which are to be compared with ***Revenue Dominated Cash Flow***, then the corresponding present worth amounts are to be computed and compared.
- Finally, the alternative with the maximum present worth amount should be selected as the best alternative.
- If we have some more alternatives which are to be compared with ***Cost Dominated Cash Flow***, then the corresponding present worth amounts are to be computed and compared.
- Finally, the alternative with the minimum present worth amount should be selected as the best alternative.

Present Worth Method_Example

Example_1 ABC Industry is planning to expand its production operation. It has identified three different technologies for meeting the goal. The initial investment and annual revenues with respect to each of the technologies are summarized in following table. Suggest the best technology which is to be implemented based on the present worth method of comparison assuming 20% interest rate, compounded annually.

Alternative	Initial Investment (Rs.)	Yearly Revenue (Rs)	Life (Years)
<i>Technology_1</i>	12,00,000	4,00,000	10
<i>Technology_2</i>	20,00,000	6,00,000	10
<i>Technology_3</i>	18,00,000	5,00,000	10

Present Worth Method_Example

Solution,

We need to suggest the best technology based on the present worth method of comparison assuming 20% interest rate, compounded annually.

For this we will calculate the present worth value of each technology and interpret the best alternative among the three.

For Technology_1

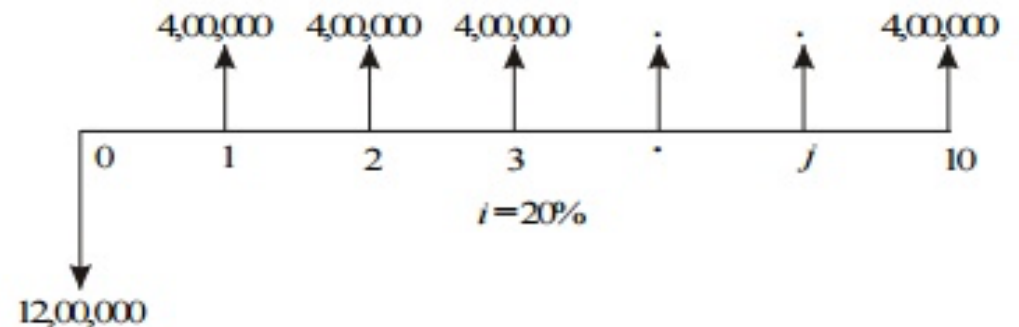
Initial Investment, $P = \text{Rs. } 12,00,000$

Annual revenue, $A = \text{Rs. } 4,00,000$

Interest rate, $i = 20\%$,

Life = 10 years

The cash flow diagram is



Present Worth Method_Example

Now,

$$PW(20\%) =$$

$$-P + R_1[1/(1+i)^1] + R_2[1/(1+i)^2] + \dots + R_j/[1/(1+i)^j] + R_n[1/(1+i)] + S[1/(1+i)^n]$$

$$= -12,00,000 + 4,00,000 (P/A, 20\%, 10)$$

$$= -12,00,000 + 4,00,000 * 4.1925$$

$$= -12,00,000 + 16,77,000$$

$$= \mathbf{4,77,000}$$

Present Worth Method_Example

For Technology_2

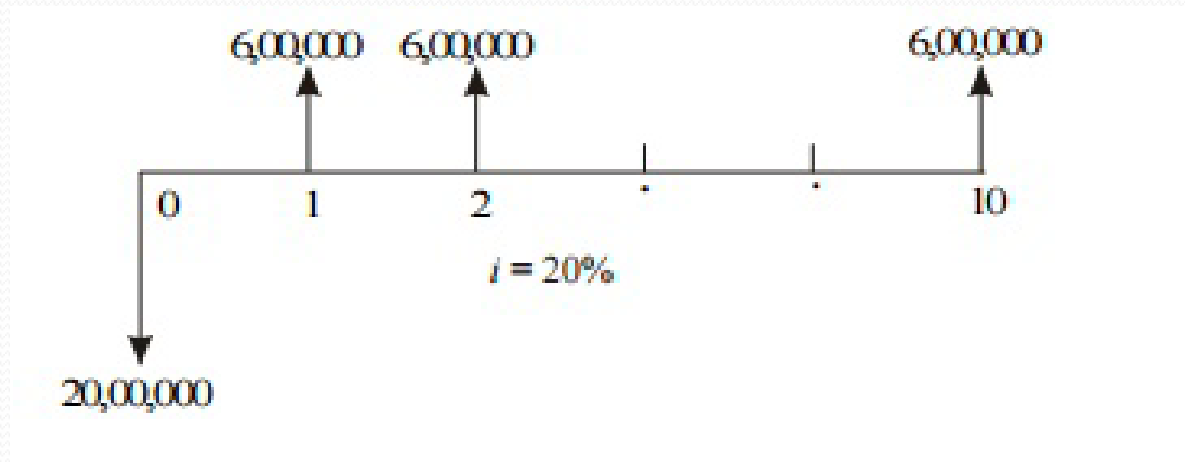
Initial Investment, $P = \text{Rs. } 20,00,000$

Annual revenue, $A = \text{Rs. } 6,00,000$

Interest rate, $i = 20\%$,

Life = 10 years

The cash flow diagram is



Present Worth Method_Example

Again,

$PW(20\%) =$

$$-P + R_1[1/(1+i)^1] + R_2[1/(1+i)^2] + \dots + R_j/[1/(1+i)^j] + R_n[1/(1+i)] + S[1/(1+i)^n]$$

$$= -20,00,000 + 6,00,000 (P/A, 20\%, 10)$$

$$= -20,00,000 + 6,00,000 * 4.1925$$

$$= -20,00,000 + 25,15,500$$

$$= \mathbf{5,15,500}$$

Present Worth Method_Example

For Technology_3

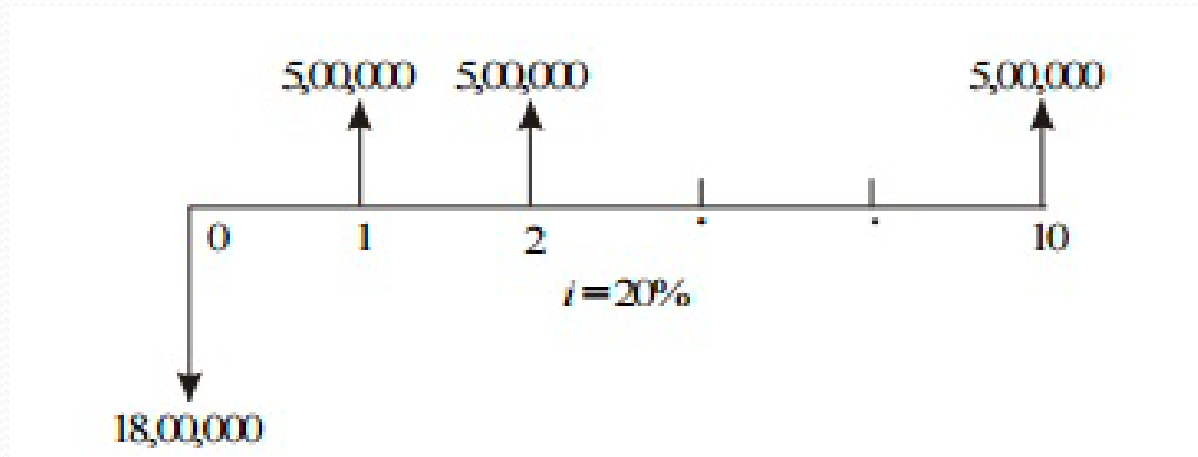
Initial Investment, $P = \text{Rs. } 18,00,000$

Annual revenue, $A = \text{Rs. } 5,00,000$

Interest rate, $i = 20\%$,

Life = 10 years

The cash flow diagram is



Present Worth Method_Example

Similarly,

$PW(20\%) =$

$$\begin{aligned} & -P + R_1[1/(1+i)^1] + R_2[1/(1+i)^2] + \dots + R_j/[1/(1+i)^j] + R_n[1/(1+i)] + S[1/(1+i)^n] \\ & = -18,00,000 + 5,00,000 (P/A, 20\%, 10) \\ & = -18,00,000 + 5,00,000 * 4.1925 \\ & = -18,00,000 + 20,96,250 \\ & = 2,96,250 \end{aligned}$$

From the above calculations, it is clear that the present worth of technology_2 is the highest among all three technologies.

Therefore, technology_2 is suggested for implementation to expand the production.

Present Worth Method_Example

Example_2 An engineer has two bids for an elevator to be installed in a new building. The details of the bids for the elevators are as follows. Determine which bid should be accepted, based on the present worth method of comparison assuming 15% interest rate, compounded annually.

Alternative	Initial Investment (Rs.)	Anual Maintenance Cost (Rs)	Life (Years)
ABC Company	4,50,000	27,000	15
XYZ Company	5,40,000	28,500	15

Present Worth Method_Example

Solution,

We need to suggest the best company based on the present worth method of comparison assuming 15% interest rate, compounded annually.

For this we will calculate the present worth value of each technology and interpret the best alternative among the three.

For ABC Company

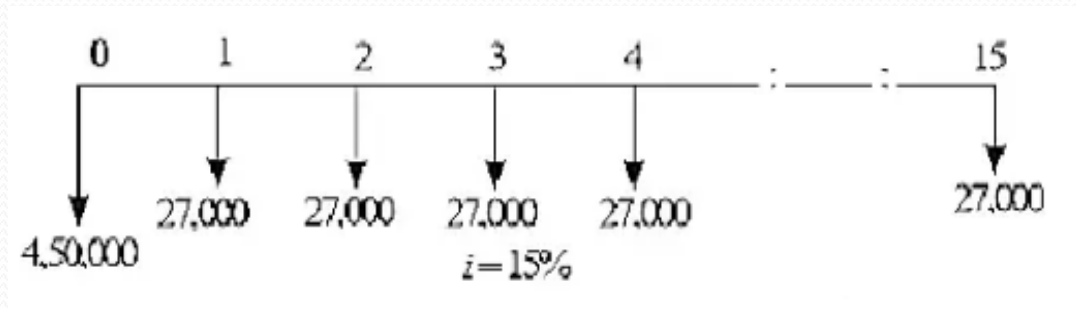
Initial cost, $P = \text{Rs. } 4,50,000$

Annual maintenance cost = Rs. 27,000

Life = 15 years

Interest rate, $i = 15\%$

The cash flow diagram is



Present Worth Method_Example

Now,

$$\begin{aligned}PW(15\%) &= 4,50,000 + 27,000(P/A, 15\%, 15) \\&= 4,50,000 + 27,000 * 5.8474 \\&= 4,50,000 + 1,57,879.80 \\&= \mathbf{6,07,879.80}\end{aligned}$$

Present Worth Method_Example

For XYZ Company

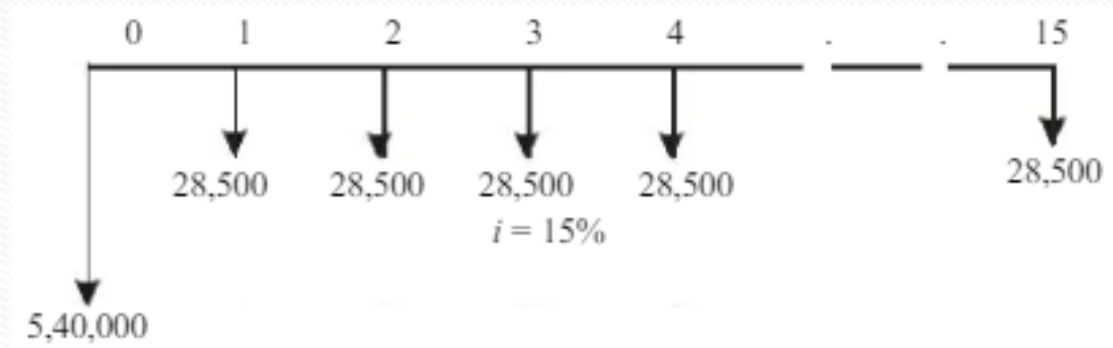
Initial cost, $P = \text{Rs. } 5,40,000$

Annual maintenance cost = Rs. 28,500

Life = 15 years

Interest rate, $i = 15\%$

The cash flow diagram is



Present Worth Method_Example

Again,

$$\begin{aligned}PW(15\%) &= 5,40,000 + 28,500(P/A, 15\%, 15) \\&= 5,40,000 + 28,500 \times 5.8474 \\&= 5,40,000 + 1,66,650.90 \\&= \mathbf{7,06,650.90}\end{aligned}$$

*The total present worth cost of bid_1 is less than that of bid_2
Hence, bid_1 is to be selected for implementation.*

*That is, the elevator from ABC Company is to be purchased
and installed in the new building.*

Present Worth Method_Example

- **Example_3** Investment proposals A and B have the net cash flows as follows:

<i>Proposal</i>	<i>End of years</i>				
	0	1	2	3	4
A (Rs.)	-10,000	3,000	3,000	7,000	6,000
B (Rs.)	-10,000	6,000	6,000	3,000	3,000

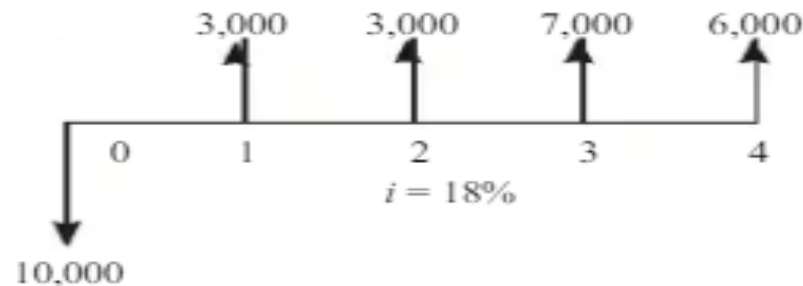
Compare the present worth of A with that of B at $i = 18\%$. Which proposal should be selected?

Present Worth Method_Example

Solution,

For Proposal A

The cash flow diagram is



$$\begin{aligned} PW_A(18\%) &= -10,000 + 3,000(P/F, 18\%, 1) + 3,000(P/F, 18\%, 2) \\ &\quad + 7,000(P/F, 18\%, 3) + 6,000(P/F, 18\%, 4) \\ &= -10,000 + 3,000(0.8475) + 3,000(0.7182) \\ &\quad + 7,000(0.6086) + 6,000(0.5158) \\ &= \mathbf{2,052.10} \end{aligned}$$

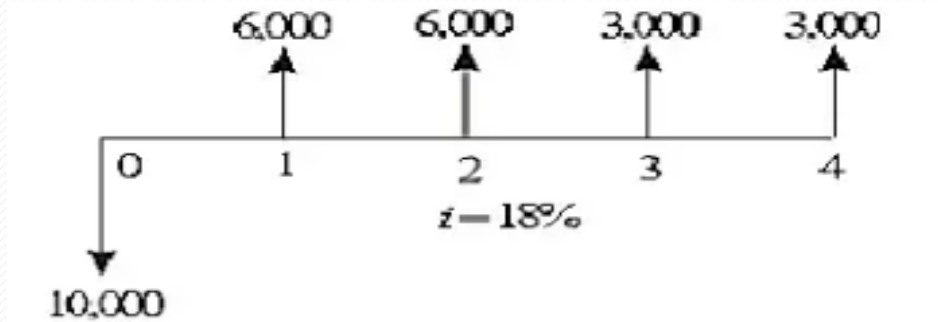
Here we have to use ***P/F*** instead of ***P/A*** because the annual revenue is not uniform

Present Worth Method_Example

Solution,

For Proposal B

The cash flow diagram is



$$\begin{aligned} PW_B(18\%) &= -10,000 + 6,000(P/F, 18\%, 1) + 6,000(P/F, 18\%, 2) \\ &\quad + 3,000(P/F, 18\%, 3) + 3,000(P/F, 18\%, 4) \\ &= -10,000 + 6,000(0.8475) + 6,000(0.7182) \\ &\quad + 3,000(0.6086) + 3,000(0.5158) \\ &= 2,767.40 \end{aligned}$$

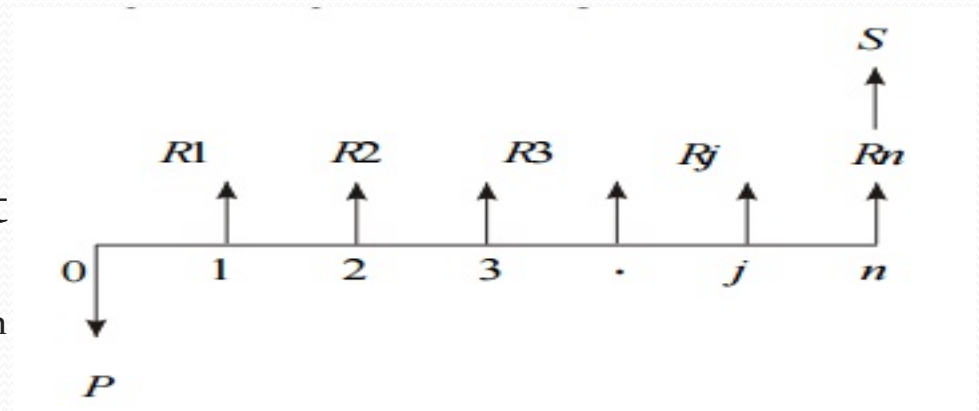
Here we have to use ***P/F*** instead of ***P/A*** because the annual revenue is not uniform

At $i = 18\%$, the present worth of proposal B is higher than that of proposal A. Therefore, select proposal B.

Future Worth Method

- In the future worth method of comparison of alternatives, the future worth of various alternatives will be computed.
- Then, the alternative with the maximum future worth of net revenue or with the minimum future worth of net cost will be selected as the best alternative for implementation.

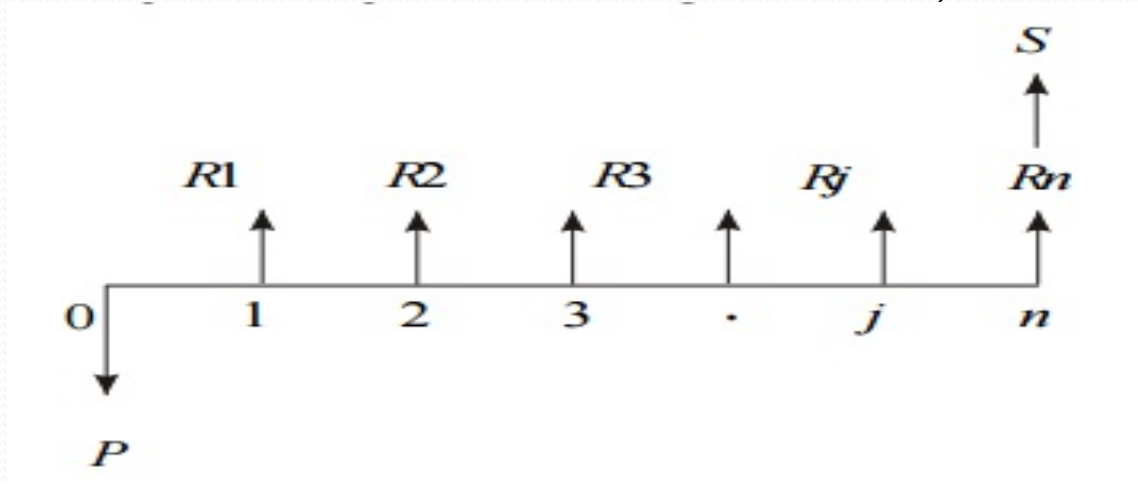
Here P represents an initial investment, R_j the net-revenue at the end of the j^{th} year, and S the salvage value at the end of the n^{th} year.



Future Worth Method

- To find the future worth of the revenue dominated cash flow diagram for a given interest rate, the formula is

$$FW(i) = -P + R_1[1/(1+i)^1] + R_2[1/(1+i)^2] + \dots + R_j/[1/(1+i)^j] + R_n + S$$



Here P represents an initial investment, R_j the net-revenue at the end of the j^{th} year, and S the salvage value at the end of the n^{th} year.

Future Worth Method

- In the above formula, the expenditure is assigned with negative sign and the revenues are assigned with positive sign.
- If we have some more alternatives which are to be compared with this alternative, then the corresponding future worth amounts are to be computed and compared.
- Finally, the alternative with the maximum future worth amount should be selected as the best alternative.

Future Worth Method_Example

Example_1 Consider the following two mutually exclusive alternatives:

Alternative	Year_0	Year_1	Year_2	Year_3	Year_4
<i>A</i>	-50,00,000	20,00,000	20,00,000	20,00,000	20,00,000
<i>B</i>	-45,00,000	18,00,000	18,00,000	18,00,000	18,00,000

At $i = 18\%$, select the best alternative based on future worth method of comparison.

Future Worth Method_Example

Solution,

For Alternative A

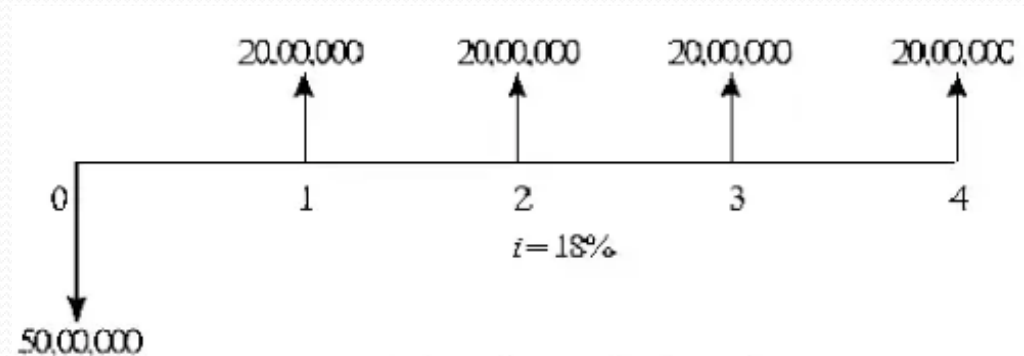
Initial investment, $P = \text{Rs. } 50,00,000$

Annual equivalent revenue, $A = \text{Rs. } 20,00,000$

Interest rate, $i = 18\%$, compounded annually

Life of alternative A = 4 years

The cash flow diagram is



Future Worth Method_Example

Now,

$$\begin{aligned}FW_A(18\%) &= -50,00,000(F/P, 18\%, 4) + 20,00,000(F/A, 18\%, 4) \\&= -50,00,000(1.939) + 20,00,000(5.215) \\&= \mathbf{7,35,000}\end{aligned}$$

For Alternative B

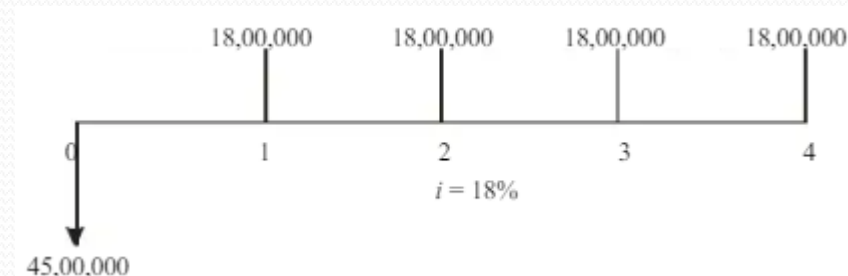
Initial investment, $P = \text{Rs. } 45,00,000$

Annual equivalent revenue, $A = \text{Rs. } 18,00,000$

Interest rate, $i = 18\%$, compounded annually

Life of alternative A = 4 years

Cash Flow diagram is:



Future Worth Method_Example

Again,

$$\begin{aligned}FW_B(18\%) &= -45,00,000(F/P, 18\%, 4) + 18,00,000(F/A, 18\%, 4) \\&= -45,00,000(1.939) + 18,00,000(5.215) \\&= \mathbf{6,61,500}\end{aligned}$$

The future worth of alternative A is greater than that of alternative B.

Thus, alternative A should be selected.

Future Worth Method_Example

Example_2 A man owns a corner plot. He must decide which of the several alternatives to select in trying to obtain a desirable return on his investment. After much study and calculation, he decides that the two best alternatives are as given in the following table:

Alternative	Build Gas Station	Build Ice Cream Stand
Initial Investment	20,00,000	36,00,000
Annual Property Tax	80,000	1,50,000
Annual Income	8,00,000	9,80,000
Life of Building (Years)	20	20
Salvage Value	0	0

Evaluate the best alternative based on future worth method of comparison at $i = 12\%$.

Future Worth Method_Example

Solution,

For Alternative_1 Build Gas Station

Initial Investment=Rs. 20,00,000

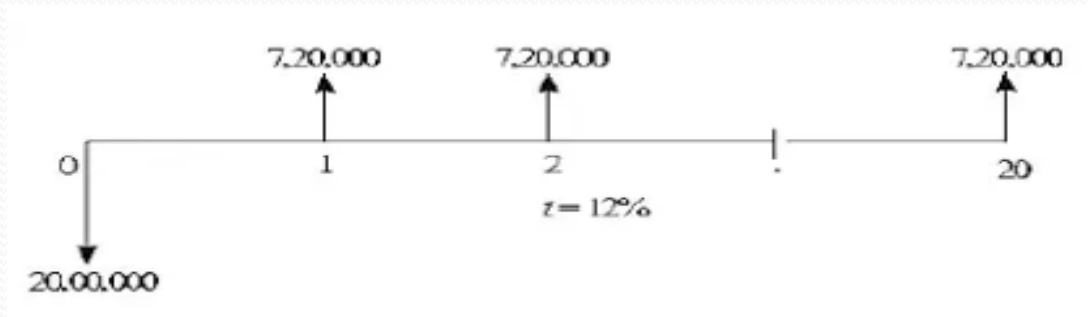
Net annual income = Annual income - Annual property tax
= Rs. 8,00,000 - Rs. 80,000
= Rs. 7,20,000

Life = 20 years

Interest rate = 12%, compounded annually

Future Worth Method_Example

The cash flow diagram for this alternative is:



$$\begin{aligned} FW(12\%) &= -20,00,000(F/P, 12\%, 20) + 7,20,000(F/A, 12\%, 20) \\ &= -20,00,000 * 9.646 + 7,20,000 * 72.052 \\ &= \text{Rs. } 3,25,85,440 \end{aligned}$$

Future Worth Method_Example

Solution,

For Alternative_2 Build soft ice-cream stand

Initial Investment=Rs. 36,00,000

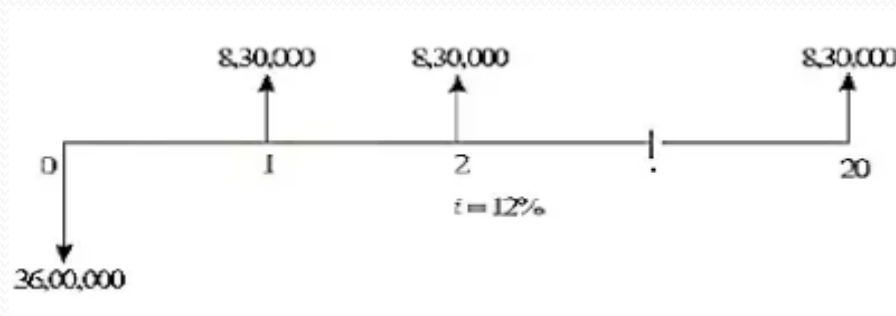
Net annual income = Annual income - Annual property tax
= Rs. 9,80,000 - Rs. 1,50,000
= Rs. 8,30,000

Life = 20 years

Interest rate = 12%, compounded annually

Future Worth Method_Example

The cash flow diagram for this alternative is:



$$\begin{aligned}FW(12\%) &= -36,00,000(F/P, 12\%, 20) + 8,30,000(F/A, 12\%, 20) \\&= -36,00,000 * 9.646 + 7,20,000 * 72.052 \\&= \text{Rs. } 2,50,77,560\end{aligned}$$

The future worth of alternative 1 is greater than that of alternative 2.

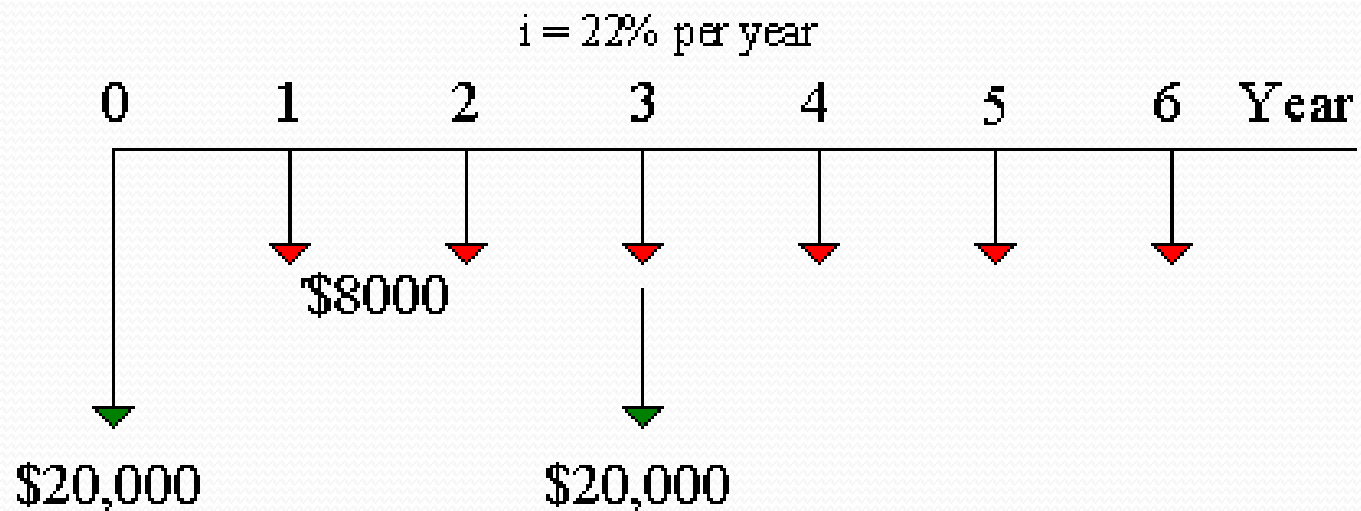
Thus, building the gas station is the best alternative.

Annual Worth Method

- Annual Worth means that all incomes and expenditures (irregular and un-uniform) are converted into an equivalent uniform annual amount, which is the same each period
- AW value of the alternatives are calculated for one life cycle only.
- If the project is continued for more than one cycle, the equivalent annual worth for the next cycle and all succeeding cycles will be exactly the same.
- It is because the provided cash flows are same for each cycle in constant-value

Annual Worth Method

- The repeatability of the uniform annual series through various life cycles can be demonstrated by considering the following cash-flow diagram which represents two life cycles of an asset with a first cost of \$20,000, an annual operating cost of \$8000, and a 3-year life.



Annual Worth Method

The AW for first life cycle (i.e., 3 years) would be calculated as follows:

$$\begin{aligned} AW &= -20,000 (A/P, 22\%, 3) - 8000 \\ &= -\$17,793 \end{aligned}$$

The AW for second life cycles is calculated as

$$\begin{aligned} AW &= -20,000(A/P, 22\%, 6) \\ &\quad -(20,000)(P/F, 22\%, 3)(A/P, 22\%, 6) \\ &\quad -8000 \\ &= -\$17,793 \end{aligned}$$

Annual Worth Method

- The AW value for the first life cycle is exactly the same as that for two life cycles.
- This same AW will be obtained when three, four, or any other number of life cycles are evaluated.
- Thus, the AW for one life cycle of an alternative represents the equivalent uniform annual worth of that alternative every time the cycle is repeated.
- When an asset has a terminal salvage value, then AW can be calculated as:

$$AW = -P(A/P, i, n) + S(A/F, i, n)$$

Annual Worth Method_Example

Example_1: Company ABC plans to purchase new equipment to improve productivity. The equipment cost is \$25000 and is expected to have a market value of \$5000 at the end of its 5-year life. If the expected improvement in productivity will net \$8000 per year and Company's MARR is 20% per year, should the Company purchase this equipment?

Solution,

Initial investment: \$25000

Salvage Value= \$5000

Annual Productivity: \$8000

Time: 5 years

Rate of Return: 20%

Annual Worth Method_Example

$AW (20\%)$

$$\begin{aligned} &= -25000(A/P, 20\%, 5) + 8000 + 5000(A/F, 20\%, 5) \\ &= -25000 * 0.33438 + 8000 + 5000 * 0.13438 \\ &= 312.40 \end{aligned}$$

Since $AW > 0$, then the equipment should be purchased.

Annual Worth Method_Example

Example_1: Projects that have the following costs are under consideration:

	<i>Project A</i>	<i>Project B</i>
<i>First cost</i>	62000	77000
<i>Annual Operating Costs</i>	15000	21000
<i>Salvage Value</i>	8000	10000
<i>Life, years</i>	4	6

Using an interest rate of 15% per year, determine which alternative should be selected on the basis of Annual Worth Analysis.

Annual Worth Method_Example

Solution,

We only need to consider one cycle of cash flow to calculate AW values:

For Project A:

First Cost: 63,000

Salvage Value= 8,000

Annual Operating Cost: 15,000

Time: 4 years

Rate of Interest: 15%

Annual Worth Method_Example

$AW_Project_A(15\%)$

$$= -62,000(A/P, 15\%, 4) - 15,000 + 8,000(A/F, 15\%, 4)$$

$$= -62,000 * 0.35027 - 15,000 + 8,000 * 0.20027$$

$$= -35,114.58$$

For Project B:

First Cost: 77,000

Salvage Value= 10,000

Annual Operating Cost: 21,000

Time: 6 years

Rate of Interest: 15%

Annual Worth Method_Example

$AW_Project_B(15\%)$

$$= -77,000(A/P, 15\%, 6) - 21,000 + 10,000(A/F, 15\%, 6)$$

$$= -77,000 * 0.26424 - 21,000 + 10,000 * 0.11424$$

$$= -40,204.08$$

On comparing the Annual Worth we select A because its AW value is numerically larger.

Internal Rate of Return (IRR)

- The internal rate of return (IRR) is a discounting cash flow technique which gives a rate of return earned by a project.
- The internal rate of return is the discounting rate where the total of initial cash outlay and discounted cash inflows are equal to zero.
- “It is the break even interest rate which equates the present worth of a project’s cash outflows to the present worth of cash inflows.”

$$NPW = PW \text{ of benefits} - PW \text{ of costs} = 0$$

- It can be used as financial metric to evaluate the performance of an investment or project.

Internal Rate of Return (IRR)

- IRR is commonly used to compare the profitability of different Investments or projects and to determine the expected return on an investment
- A higher value of it's IRR indicates that an investment or project is more profitable.
- IRR can be calculated using:

$$IRR = \frac{\text{Cash flows}}{(1+r)^i} - \text{Initial Investment}$$

Where:

Cash flows= Cash flows in the time period

r = Discount rate

i = Time period

Internal Rate of Return (IRR)

- In real life scenarios, since the investment in any project will be huge and will have a long-term effect, an organization uses a combination of various techniques of capital budgeting like NPV, IRR and payback period to select the best project.

Benefit-Cost Ratio(BCR)

- Benefit-Cost Ratio” refers to the financial metric that helps in assessing the viability of an upcoming project based on its expected costs and benefits.
- In other words, the ratio determines the relationship between the expected incremental benefit from a project and the corresponding costs that would be incurred to complete the project.
- Benefit-cost ratio (BCR) analysis is a method used to determine the economic feasibility of a project or investment.

Benefit-Cost Ratio(BCR)

- It involves comparing the total benefits of a project to its total costs, and expressing the result as a ratio.
- A BCR greater than 1 indicates that the project with good investment and BCR less than 1 indicates that the project with poor investment.

$$BCR = \frac{\textit{Present Value of Expected Benefit}}{\textit{Present Value of Expected Cost}}$$

Benefit-Cost Ratio(BCR)

- **Example:** Consider a company that has recently invested 10,000 for the purpose of replacing some of its machinery components. This renovation is expected to result in incremental benefits of 5,000 in 1st year, 3,000 in 2nd year and 4,000 in 3rd year. Calculate the benefit-cost ratio of the replacement project if the applicable discounting rate is 5%.

Benefit-Cost Ratio(BCR)

Solution,

Here, Initial Investment: 10,000

Benefit in 1st Year: 5,000

Benefit in 2nd Year: 3,000

Benefit in 3rd Year: 4,000

Discounting rate: 5%

Now,

PV of benefit in 1st year = $5,000 / (1 + 5\%)^1 = 4,761.90$

PV of benefit in 2nd year = $3,000 / (1 + 5\%)^2 = 2,721.09$

PV of benefit in 3rd year = $4,000 / (1 + 5\%)^3 = 3,455.35$

Benefit-Cost Ratio(BCR)

$$\begin{aligned} PV \text{ of Expected Benefits} &= 4,761.90 + 2,721.09 + 3,455.35 \\ &= \mathbf{10,938.34} \end{aligned}$$

Then,

$$\begin{aligned} BCR &= \frac{\text{Present Value of Expected Benefit}}{\text{Present Value of Expected Cost}} \\ &= \frac{10,938.34}{10,000} \\ &= \mathbf{1.09} \end{aligned}$$

*Hence, the benefit-cost ratio of the project is 1.09.
This indicates that it will create additional value and as such it should be considered positively.*

Uniform Gradient Uniform Series

- It calculates the BCR by assuming that the benefits and costs of the project will be received or incurred at a uniform rate over time.
- It is a cash flow where the increases or decreases by some amount in each cash flow period.
- **Example:** If cash flow gradient in *period 1* is \$500, then in *period 2*, it will be \$600 and so on with amount increasing by \$100 in each periods.
- This is a gradient cash flow with the gradient (G) equals to \$100.

Mutually Exclusive and Independent Alternatives

- When alternatives are mutually exclusive only one can be selected because only one is needed.
- Therefore, the alternatives are *compared against each other* and the one having the most favorable PW is identified as “the best”.
- For example, in evaluating locations for construction of a new manufacturing facility, only one site would be selected because only one is needed.

Mutually Exclusive and Independent Alternatives

- As soon as the best one is identified, all of the others are automatically excluded.
- All alternatives which yield a return of at least the minimum acceptable rate of return (MARR) are accepted.
- In terms of a present worth analysis, all alternatives which have a PW equal to or greater than 0 when i is equal to the MARR would be considered as acceptable alternatives.

Mutually Exclusive and Independent Alternatives

- When comparing mutually exclusive alternatives, BCR is also used to compare the relative feasibility of each alternative.
- This would be done by calculating the BCR for each alternative and comparing the results, with the highest BCR is considered most economically feasible.

Numerical_1

Calculate the discounted payback period for the given two projects and state which one is worthwhile and why?

Year	Project_1 (Cash Flow)	Project_2 (Cash Flow)
0	-80,000	-70,000
1	30,000	60,000
2	40,000	20,000
4	60,000	40,000
5	40,000	40,000

Numerical_1

Solution,

Since discount rate is not given in question.

Let us take discount rate as 10% for both projects.

Now, we can calculate discount for Project_1 and Project_2 using formula

$$\text{Discount} = \frac{\text{Cash Flow}}{(1+r)^n}$$

where, r is the discount rate and n is year.

Numerical_1

For Project_1

$$\begin{aligned}\text{Year}_0 (\text{discount}) &= \frac{\text{Cash Flow}}{(1+r)^n} \\ &= \frac{-80,000}{(1+0.1)^0} \\ &= -80,000\end{aligned}$$

$$\begin{aligned}\text{Year}_1 (\text{discount}) &= \frac{\text{Cash Flow}}{(1+r)^n} \\ &= \frac{30,000}{(1+0.1)^1} \\ &= 27,272.72\end{aligned}$$

$$\begin{aligned}\text{Year}_2 (\text{discount}) &= \frac{\text{Cash Flow}}{(1+r)^n} \\ &= \frac{40,000}{(1+0.1)^2} \\ &= 33,057.85\end{aligned}$$

$$\begin{aligned}\text{Year}_4 (\text{discount}) &= \frac{\text{Cash Flow}}{(1+r)^n} \\ &= \frac{60,000}{(1+0.1)^4} \\ &= 40,980.80\end{aligned}$$

Numerical_1

$$\begin{aligned} \text{Year}_5 (\text{discount}) &= \frac{\text{Cash Flow}}{(1+r)^n} \\ &= \frac{40,000}{(1+0.1)^5} \\ &= 24,836.85 \end{aligned}$$

Here, Year₃ is not given so, we should calculate Year 3 as:

$$\begin{aligned} \text{Year}_3 \text{ cash flow} &= \text{Cash flow of Year}_2 + \frac{\text{Change in Cash Flow}}{\text{No.of Years between } -1} \\ &= 40,000 + \frac{4^{\text{th}} \text{ Year Cash Flow} - 2^{\text{nd}} \text{ Year Cash Flow}}{2 - 1} \\ &= 40,000 + 60,000 - 40,000 \\ &= 60,000 \end{aligned}$$

Numerical_1

So,

$$\begin{aligned} \text{Year}_3 (\text{discount}) &= \frac{\text{Cash Flow}}{(1+r)^n} \\ &= \frac{60,000}{(1+0.1)^3} \\ &= 45,078.88 \end{aligned}$$

Year	Cash Flow	Discount	Cumulative Discount
0	-80,000	-80,000	-80,000
1	30,000	27,272.72	-52,727.28
2	40,000	33,057.85	-19,669.43
3	60,000	45,078.88	25,409.45
4	60,000	40,980.80	66,390.25
5	40,000	24,836.85	91,227.1

Numerical_1

*Since, cumulative discount turns positive in 3rd year discounted
Payback period for Project_1.*

So, discounted payback period is:

$$\begin{aligned} & \text{Previous Year} + \frac{\text{Cumulative discount positive value of Previous Years}}{\text{Discount of first positive year}} \\ &= 2 + \frac{19,669.43}{45,078.88} \\ &= 2 + 0.436 \\ &= 2.436 \text{ years} \end{aligned}$$

Numerical_1

For Project_2

$$\begin{aligned}\text{Year}_0 (\text{discount}) &= \frac{\text{Cash Flow}}{(1+r)^n} \\ &= \frac{-70,000}{(1+0.1)^0} \\ &= -70,000\end{aligned}$$

$$\begin{aligned}\text{Year}_1 (\text{discount}) &= \frac{\text{Cash Flow}}{(1+r)^n} \\ &= \frac{60,000}{(1+0.1)^1} \\ &= 54,545.45\end{aligned}$$

$$\begin{aligned}\text{Year}_2 (\text{discount}) &= \frac{\text{Cash Flow}}{(1+r)^n} \\ &= \frac{20,000}{(1+0.1)^2} \\ &= 16,528.92\end{aligned}$$

$$\begin{aligned}\text{Year}_4 (\text{discount}) &= \frac{\text{Cash Flow}}{(1+r)^n} \\ &= \frac{40,000}{(1+0.1)^4} \\ &= 27,320.53\end{aligned}$$

Numerical_1

$$\begin{aligned} \text{Year}_5 (\text{discount}) &= \frac{\text{Cash Flow}}{(1+r)^n} \\ &= \frac{40,000}{(1+0.1)^5} \\ &= 24,836.85 \end{aligned}$$

Here, Year₃ is not given so, we should calculate Year 3 as:

$$\begin{aligned} \text{Year}_3 \text{ cash flow} &= \text{Cash flow of Year}_2 + \frac{\text{Change in Cash Flow}}{\text{No.of Years between } -1} \\ &= 20,000 + \frac{4^{\text{th}} \text{ Year Cash Flow} - 2^{\text{nd}} \text{ Year Cash Flow}}{2 - 1} \\ &= 20,000 + 40,000 - 20,000 \\ &= 40,000 \end{aligned}$$

Numerical_1

So,

$$\begin{aligned} \text{Year}_3 (\text{discount}) &= \frac{\text{Cash Flow}}{(1+r)^n} \\ &= \frac{40,000}{(1+0.1)^3} \\ &= 30,052.59 \end{aligned}$$

Year	Cash Flow	Discount	Cumulative Discount
0	-70,000	-70,000	-70,000
1	60,000	54,545.45	-15,454.55
2	20,000	16,528.92	1,074.37
3	40,000	30,052.59	31,126.96
4	40,000	27,320.53	58,447.49
5	40,000	24,836.85	83,284.34

Numerical_1

Since, cumulative discount turns positive in 2nd year discounted Payback period for Project_2.

So, discounted payback period is:

$$\begin{aligned} &= \text{Previous Year} + \frac{\text{Cumulative discount positive value of Previous Years}}{\text{Discount of first positive year}} \\ &= 1 + \frac{15,454.55}{16,528.92} \\ &= 1 + 0.935 \\ &= 1.935 \text{ years} \end{aligned}$$

Since discounted payback period for Project_1 is 2.436 years and Project_2 is 1.935 years.

Hence Project_2 will be more worthwhile because we will recover initial cost faster than Project_1 which will lead to more profit than Project_1.

Numerical_2

Which project will you choose from the given data and why?

Year	Project_A	Project_B
0	-40,000	-80,000
1	5,000	2,000
2	15,000	10,000
3	5,000	70,000
4	40,000	50,000
5	50,000	50,000

Numerical_2

Solution,

Since discount rate is not given in question.

Let us take discount rate as 10% for both projects.

Now, we can calculate discount for Project_A and Project_B using formula

$$\text{Discount} = \frac{\text{Cash Flow}}{(1+r)^n}$$

where, r is the discount rate and n is year.

Numerical_2

For Project_A

$$\begin{aligned} \text{Year}_0 (\text{discount}) &= \frac{\text{Cash Flow}}{(1+r)^n} \\ &= \frac{-40,000}{(1+0.1)^0} \\ &= -40,000 \end{aligned}$$

$$\begin{aligned} \text{Year}_1 (\text{discount}) &= \frac{\text{Cash Flow}}{(1+r)^n} \\ &= \frac{5,000}{(1+0.1)^1} \\ &= 4,545.45 \end{aligned}$$

$$\begin{aligned} \text{Year}_2 (\text{discount}) &= \frac{\text{Cash Flow}}{(1+r)^n} \\ &= \frac{15,000}{(1+0.1)^2} \\ &= 12,396.69 \end{aligned}$$

$$\begin{aligned} \text{Year}_3 (\text{discount}) &= \frac{\text{Cash Flow}}{(1+r)^n} \\ &= \frac{5,000}{(1+0.1)^3} \\ &= 3,756.57 \end{aligned}$$

Numerical_2

$$\begin{aligned} \text{Year}_4 (\text{discount}) &= \frac{\text{Cash Flow}}{(1+r)^n} \\ &= \frac{40,000}{(1+0.1)^4} \\ &= 27,320.53 \end{aligned}$$

$$\begin{aligned} \text{Year}_5 (\text{discount}) &= \frac{\text{Cash Flow}}{(1+r)^n} \\ &= \frac{50,000}{(1+0.1)^5} \\ &= 31,046.06 \end{aligned}$$

Numerical_2

Year	Cash Flow	Discount	Cumulative Discount
0	-40,000	-40,000	-40,000
1	5,000	4,545.45	-35,454.55
2	15,000	12,369.69	-23,084.86
3	5,000	3,756.57	-19,328.29
4	40,000	27,320.53	8,011.24
5	50,000	31,046.06	39,057.3

Numerical_2

*Since, cumulative discount turns positive in 4th year discounted
Payback period for Project_A.*

So, discounted payback period is:

$$\begin{aligned} & \text{Previous Year} + \frac{\text{Cumulative discount positive value of Previous Years}}{\text{Discount of first positive year}} \\ &= 3 + \frac{19,301.29}{27,320.53} \\ &= 3 + 0.706 \\ &= 3.706 \text{ years} \end{aligned}$$

Numerical_2

For Project_B

$$\begin{aligned}\text{Year}_0 (\text{discount}) &= \frac{\text{Cash Flow}}{(1+r)^n} \\ &= \frac{-80,000}{(1+0.1)^0} \\ &= -80,000\end{aligned}$$

$$\begin{aligned}\text{Year}_1 (\text{discount}) &= \frac{\text{Cash Flow}}{(1+r)^n} \\ &= \frac{2,000}{(1+0.1)^1} \\ &= 1,818.18\end{aligned}$$

$$\begin{aligned}\text{Year}_2 (\text{discount}) &= \frac{\text{Cash Flow}}{(1+r)^n} \\ &= \frac{10,000}{(1+0.1)^2} \\ &= 8,264.46\end{aligned}$$

$$\begin{aligned}\text{Year}_3 (\text{discount}) &= \frac{\text{Cash Flow}}{(1+r)^n} \\ &= \frac{70,000}{(1+0.1)^3} \\ &= 52,592.03\end{aligned}$$

Numerical_2

$$\begin{aligned} \text{Year}_4 (\text{discount}) &= \frac{\text{Cash Flow}}{(1+r)^n} \\ &= \frac{50,000}{(1+0.1)^4} \\ &= 34,150.67 \end{aligned}$$

$$\begin{aligned} \text{Year}_5 (\text{discount}) &= \frac{\text{Cash Flow}}{(1+r)^n} \\ &= \frac{50,000}{(1+0.1)^5} \\ &= 31,046.06 \end{aligned}$$

Numerical_2

Year	Cash Flow	Discount	Cumulative Discount
0	-80,000	-80,000	-80,000
1	2,000	1,818.18	-78,181.82
2	10,000	8,264.46	-69,917.36
3	70,000	52,592.03	-17,325.33
4	50,000	34,150.67	16,825.34
5	50,000	31,046.06	47,871.40

Numerical_2

Since, cumulative discount turns positive in 4th year discounted Payback period for Project_B.

So, discounted payback period is:

$$\begin{aligned} &= \text{Previous Year} + \frac{\text{Cumulative discount positive value of Previous Years}}{\text{Discount of first positive year}} \\ &= 3 + \frac{17,325.33}{34,150.67} \\ &= 3 + 0.507 \\ &= 3.507 \text{ years} \end{aligned}$$

Since, discount Payback Period for Project_A is 3.706 years and Project_B is 3.507 years. Hence Project_B will be more worthwhile because we will recover initial cost faster than Project_A which will lead to more profit than Project_A

Numerical_3

Suppose you have purchased 1,000 pieces of pants for Rs. 800 per piece. You would then sell the pants to a store for Rs. 1,000 per piece. Your expense as a transportation cost is Rs. 1500 . Calculate ***ROI (Rate of Return)***.

Numerical_3

Solution,

Here we have,

Cost price of one pant = 800

*Cost price of 1,000 pant (CP) = $800 \times 1,000$
= Rs. 8,00,000*

Transportation cost (TC) = 1,500

Selling price of one pant = 1,000

*Selling price of 1,000 Pant (SP) = $1,000 \times 1,000$
= Rs. 10,00,000*

Numerical_3

$$\begin{aligned}\text{Total investment} &= CP + TC \\ &= 8,00,000 + 1,500 \\ &= 8,01,500\end{aligned}$$

$$\begin{aligned}\text{Profit} &= SP - \text{Total investment} \\ &= 10,00,000 - 8,01,500 \\ &= 1,98,500\end{aligned}$$

$$\begin{aligned}\text{ROI} &= \frac{\text{Profit}}{\text{Total Investment}} \times 100\% \\ &= \frac{1,98,500}{8,01,500} \times 100\% \\ &= 24.76\%\end{aligned}$$

Therefore the ROI for the investment is 24.76%

Numerical_4

Calculate ROI for following project

Year	Transactions
0	-50,000
1	5,000
2	5,000
3	20,000
4	40,000
5	50,000

Assignment-I

- Differentiate between software project management and other types of project management.
- Briefly explain project management cycle.
- Explain product life cycle with example.
- What are different steps of Project Planning? Describe activities in each step.
- Define payback period. How it can be used as an evaluation technique? What are its advantages and disadvantages?
- Explain different approach of Economical Assessment for project.
- Differentiate between strategic Assessment and technical assessment. Explain the steps involved in cost benefit analysis?

Submission Deadline: 30th May_2025



End of Chapter 2

Thank You !!!!